

Q1: 16 March (Shift 1) - Numerical

The equivalents of ethylene diamine required to replace the neutral ligands from the coordination sphere of the trans-complex of $\text{CoCl}_3 \cdot 4\text{NH}_3$ is _____.

(Round off to the Nearest Integer).

Q2: 16 March (Shift 2) - Numerical

$[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ absorbs light of wavelength 498 nm during a d – d transition. The octahedral splitting energy for the above complex is $\text{_____} \times 10^{-19}$ J. (Round off to the Nearest Integer).

$$h = 6.626 \times 10^{-34} \text{ Js}; c = 3 \times 10^8 \text{ ms}^{-1}$$

Q3: 17 March (Shift 2) - Single Correct

Match List-I with List-II :

List-I	List-II
(a) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$	(i) Linkage isomerism
(b) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$	(ii) Solvate isomerism
(c) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$	(iii) Co-ordination isomerism
(d) $\text{cis-}[\text{CrCl}_2(\text{ox})_2]^{3-}$	(iv) Optical isomerism

Choose the correct answer from the options given below :

(1) (a) – (iii), (b) – (i), (c) – (ii), (d) – (iv)

(2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Q4: 18 March (Shift 1) - Single Correct

The correct structures of trans- $[\text{NiBr}_2(\text{PPh}_3)_2]$ and meridional- $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$, respectively, are



Q5: 18 March (Shift 1) - Numerical

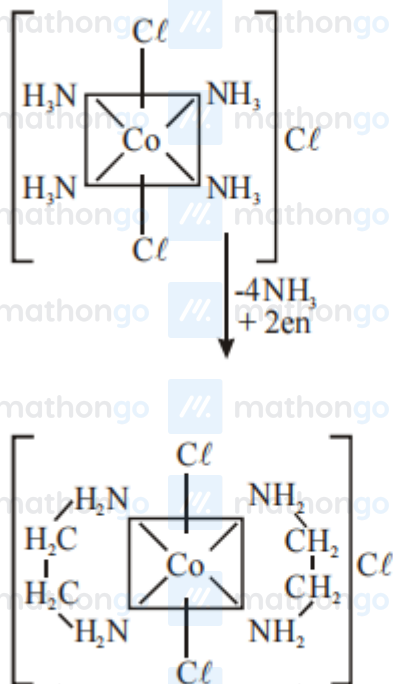
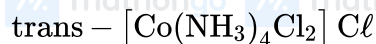
The total number of unpaired electrons present in the complex $K_3[Cr(\text{oxalate})_3]$ is ____.

Answer Key**Q1 (2)****Q2 (4)****Q3 (1)****Q4 (4)****Q5 (3)**

Q1 (2)



or



As we know that ethylene diamine is a bidentate ligand and ammonia is a monodentate ligand. It means overall two ethylene diamine is required to replace the all neutral ligands (four ammonia) from the coordination sphere of this complex.

Q2 (4)

$$\lambda_{\text{absorbed}} = 498 \text{ nm (given)}$$

The octahedral splitting energy

$$\Delta_0 \text{ or } E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{498 \times 10^{-9}}$$

$$= 0.0399 \times 10^{-17} \text{ J}$$

$$= 3.99 \times 10^{-19} \text{ J}$$

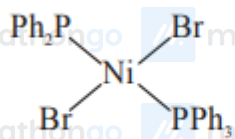
$$= 4.00 \times 10^{-19} \text{ J (round off)}$$

Q3 (1)

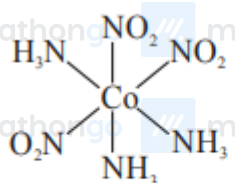
Complex	Type of Isomerism
(a) $[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$	Co-ordination isomerism
(b) $[\text{Co}(\text{NH}_3)_3 (\text{NO}_2)_3]$	Linkage isomerism
(c) $[\text{Cr}(\text{H}_2\text{O})_6] \text{Cl}_3$	Solvate isomerism
(d) $\text{cis-}[\text{CrCl}_2(\text{ox})_2]^{3-}$	Optical isomerism

Q4 (4)

$\text{trans-}[\text{Ni Br}_2(\text{PPh}_3)_2]$ is



meridional - $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$ is



Q5 (3)

$\text{K}_3 [\text{Cr}(\text{oxalate})_3]$

Chromium is in +3 oxidation state. Number of unpaired electrons in Cr^{+3} will be 3